Space technology for Earth

Although some people see space exploration as purely scientific research or even a waste of money, in reality space technology has transformed all our lives. Aside from the many benefits brought by satellites, the challenge of getting humans into space and returning them safely to Earth has inspired countless engineers to make advances that have later found applications on Earth.

SATELLITE ARCHAEOLOGY

Remote sensing has found a huge range of uses (see pp.262–263), but one of the most surprising is in archaeology. Researchers using satellite photos can identify ancient tracks and disturbances that reveal the presence of buried ruins such as a lost city, nicknamed the Atlantis of the Sands, at Shisr in Oman.

Digital camera with lens removed

DIGITAL IMAGING

Early space probes and satellites used either television cameras or photographic film to capture images. In the early 1990s, NASA scientist Eric Fossum invented an improved sensor to capture images as digital electronic data. These CMOS sensors are now widely used in cell phones, digital cameras, and other imaging systems.

SPACE BLANKETS

The insulating foil blankets that are provided after endurance races and to keep people warm in disaster areas originated from insulation used on the Apollo lunar landing module. Made by applying a thin aluminum coating to a plastic film, they reflect infrared radiation and trap heat close to the wearer's body.